

TECHNICAL PRESENTATION

Sampling & Resolution

How discrete measurement shapes the signals we capture

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The World Is Continuous. Computers Are Not.

Sound, light, temperature, motion — every real-world signal varies smoothly over time and space.

A computer, however, can only store a finite list of numbers.

To turn a continuous signal into that finite list, every digital system must answer two independent questions:

1

How often do we measure?

→ Sampling

2

How finely do we measure?

→ Resolution

Answer either one incorrectly and information is destroyed permanently — no downstream processing can bring it back.



A smooth signal (top) is captured only at discrete instants, producing a finite sequence of values (bottom).

Sampling: How Often We Measure

Sampling captures the value of a signal at discrete points in time or space, at a fixed sampling rate f_s (samples per second).

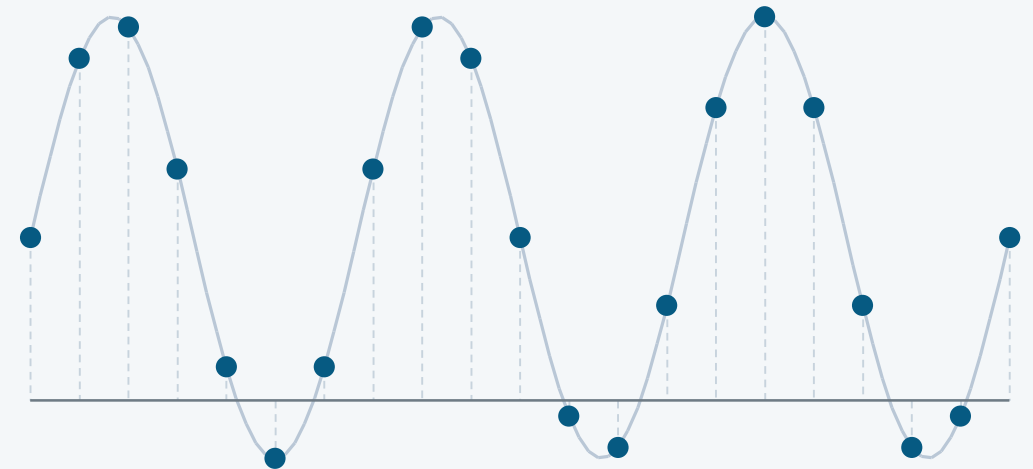
NYQUIST-SHANNON THEOREM

$$f_s \geq 2 \times f_{\max}$$

Sample too slowly and distinct frequencies become indistinguishable — an effect called **aliasing**.

Practical rates: audio CDs sample at 44.1 kHz to reconstruct sound up to ~22 kHz — safely above the range of human hearing.

SIGNAL SAMPLED AT $f_s \geq 2f_{\max}$



Enough samples per cycle preserve the exact shape of the original waveform on reconstruction.

Resolution: How Finely We Measure

Resolution is the smallest distinguishable increment in the quantity being measured — set by the number of discrete levels available.

- **Amplitude (ADC bit-depth)**

n-bit converter → 2^n quantisation levels

- **Spatial (imaging)**

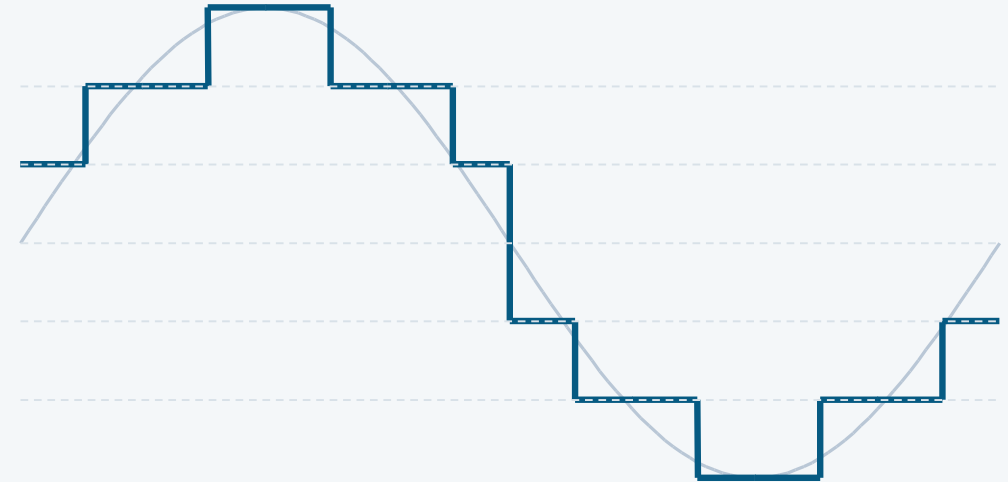
Pixels per inch (PPI) or per millimetre

- **Temporal**

Smallest resolvable time interval between events

8-bit → 256 levels 16-bit → 65,536 levels

SMOOTH SIGNAL vs QUANTISED LEVELS



Coarse quantisation (few levels) forces the true curve into a staircase — the gap between curve and steps is quantisation error.

Sampling and Resolution Are Independent — Fidelity Needs Both

Sampling sets the time/space axis; resolution sets the precision axis. Total data captured scales with both: data rate $\approx f_s \times \text{bits/sample}$.

Low sampling · Low resolution

Both signal shape and detail are lost.

Poor fidelity

Low sampling · High resolution

Fine detail on values that are already timed too coarsely to trust.

Timing-limited

High sampling · Low resolution

Timing is precise, but each reading is coarse and noisy.

Precision-limited

High sampling · High resolution

Signal shape and detail are both preserved — at the cost of data volume.

High fidelity

Verifying the Design: Analytical & Visual Tools

Frequency-domain analysis (FFT)

Confirms $f_s \geq 2f_{\max}$ and reveals aliasing as spurious low-frequency artifacts.

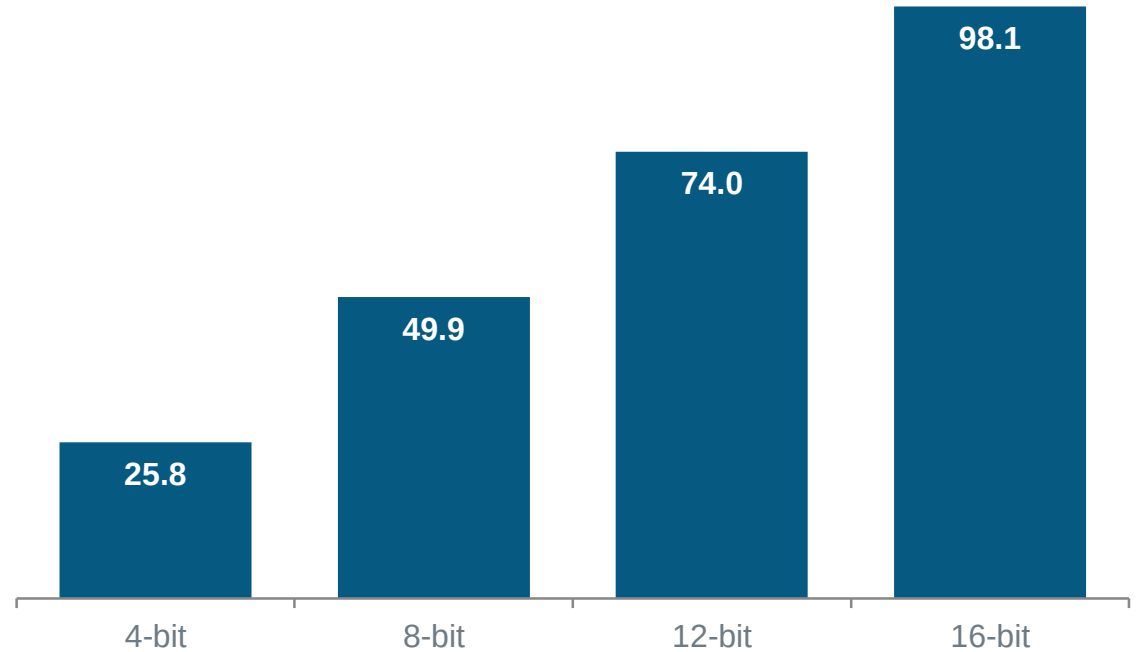
Time-domain reconstruction plots

Overlay original vs. reconstructed signal to visually confirm no shape distortion.

Quantisation SNR

$\text{SNR(dB)} \approx 6.02 \times n + 1.76$ — each added bit gains ~6 dB of signal-to-noise ratio.

QUANTISATION SNR BY BIT-DEPTH



KEY TAKEAWAYS

Two Independent Dials, One Fidelity Outcome

01 Sampling sets the timing floor

The Nyquist–Shannon theorem ($f_s \geq 2f_{\max}$) defines the minimum rate needed to avoid aliasing.

02 Resolution sets the precision ceiling

Bit-depth and quantisation levels define how finely each sample can be measured.

03 Validate with FFT and SNR analysis

Frequency-domain and error-based tools confirm whether a design choice is sufficient — before deployment.

04 Communicate the trade-off early

Storage, bandwidth, and cost all scale with $f_s \times \text{resolution}$ — a decision stakeholders should see up front.

Thank you — questions welcome.